The angular momentum of the rod and two balls about the pivot is

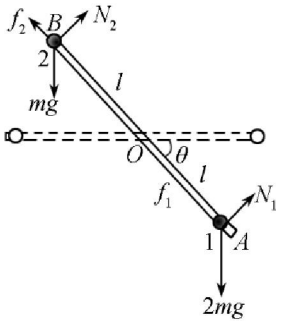
$$L = 2mlv_1 + mlv_2 , \qquad (4)$$

The angular-momentum law gives

$$2mgl\cos\theta - mgl\cos\theta = \frac{\Delta L}{\Delta t} 。 \qquad (5)$$

By the definition of angular acceleration β,

$$\beta = \frac{\Delta\omega}{\Delta t} , \qquad (6)$$



Equations (2)(4)(5)(6) give

$$\beta = \frac{g\cos\theta}{3l} 。 \qquad (7)$$

While both are on the rod, they circle the pivot. Ball 1 has the following tangential and normal accelerations:

$$a_{1t} = l\beta \qquad (8)$$

$$a_{1t} = l\omega \qquad (9)$$

Let N_1 be the force normal to the rod on ball 1, and f_1 the force on ball 1 along the rod. Newton's second law gives

$$2mg\cos\theta - N_1 = 2ma_{1t} , \qquad (10)$$

$$f_1 - 2mg\sin\theta = 2ma_{1t} \qquad (11)$$

Equations (3)(9)(10)(11) give

$$N_1 = \frac{4}{3}mg\cos\theta 。 \qquad (12)$$

$$f_1 = \frac{10}{3}mg\sin\theta 。 \qquad (13)$$

For ball 2, the force normal to the rod on ball 2 has magnitude N_2; the force along the rod on ball 2 has magnitude f_2. The same analysis gives

$$N_2 = \frac{4}{3}mg\cos\theta , \qquad (14)$$

$$f_2 = \frac{1}{3}mg\sin\theta 。 \qquad (15)$$

2 By (13)(14)(15), balls 1 and 2 have equal static-friction limits; show that ball 1 needs more friction than ball 2. During rotation, ball 1 reaches the limit first, so ball 1 slides first. As ball 1 starts to slide, let the angle